



11908 - Probing the M-sigma Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NIRSPEC 3XMM	NIRSpec Fixed Slit Spectroscopy	(2) 3XMM_J2150-05_host
	2	NIRSPEC EP	NIRSpec Fixed Slit Spectroscopy	(1) EP20240222a_host
	3	NIRCAM EP	NIRCam Imaging	(1) EP20240222a_host
	4	NIRCAM 3XMM	NIRCam Imaging	(2) 3XMM_J2150-05_host

ABSTRACT

We aim to test the extension of the black hole–host galaxy $M_{\text{BH}}-\sigma_{\text{star}}$ relation into the intermediate-mass black hole (IMBH) regime using two exceptional off-nuclear tidal disruption events (TDEs), 3XMM J2150–05 and EP20240222a. These transients, hosted by ultra-compact dwarfs, exhibit thermal X-ray spectra and luminosities consistent with accretion onto $10^4-10^5 M_{\odot}$ black holes—among the most compelling IMBH candidates known. We will use JWST/NIRSpec high-resolution spectroscopy to measure the stellar velocity dispersions of their

hosts from CO absorption features, and NIRCам multi-filter imaging to resolve their morphology, effective radii, and stellar masses. Together, these observations will provide the first direct constraints on the stellar environments of accreting IMBHs and determine whether the $M_{\text{BH}}-\sigma_{\text{star}}$ relation—central to our understanding of black hole–galaxy co-evolution—extends into the low-mass regime, while revealing the stellar conditions that foster IMBH formation and growth.

OBSERVING DESCRIPTION

We will obtain fixed-slit NIRSpec high-resolution (G235H/F170LP) spectroscopy of the compact dwarf galaxies hosting the two off-nuclear tidal disruption events. These observations will measure the stellar velocity dispersion of the compact stellar systems. The total exposure times are 3.5 hours and 6.5 hours for 3XMM~J2150–05 and EP~20240222a, respectively.

We will also acquire deep NIRCам imaging with two 20-minute exposures per target, using a total of four filters (F115W, F150W, F227W, and F444W). These data will be used to characterize the morphology, compactness, and stellar mass of the host systems.

Proposal 11908 - Targets - Probing the M-sigma Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	EP20240222a_host	RA: 11 32 6.1440 (173.0256000d) Dec: +27 00 18.00 (27.00500d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]				
Fixed Targets	(2)	3XMM_J2150-05_host	RA: 21 50 22.5000 (327.5937500d) Dec: -05 51 9.00 (-5.85250d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]				

Proposal 11908 - Observation 1 - Probing the M-sigma Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs

Observation	Proposal 11908, Observation 1: NIRSPEC 3XMM										Tue May 26 23:00:45 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnos	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	3XMM_J2150-05_host	RA: 21 50 22.5000 (327.5937500d) Dec: -05 51 9.00 (-5.85250d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 3XMM_J2150-05_host	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	274087	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			FULL					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						BOTH				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	S200A1	NRSIRS2	48	1	1	NONE	8	8	28127.38	

Proposal 11908 - Observation 2 - Probing the M-sigma Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs

Observation	Proposal 11908, Observation 2: NIRSPEC EP											Tue May 26 23:00:45 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	EP20240222a_host	RA: 11 32 6.1440 (173.0256000d) Dec: +27 00 18.00 (27.00500d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 EP20240222a_host	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	299843	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						BOTH				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	S200A1	NRSIRS2	22	1	1	NONE	8	8	12954.934	

Proposal 11908 - Observation 3 - Probing the M-sigma Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs

Observation	Proposal 11908, Observation 3: NIRCAM EP										Tue May 26 23:00:45 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCam Imaging										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	EP20240222a_host	RA: 11 32 6.1440 (173.0256000d) Dec: +27 00 18.00 (27.00500d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Compact dwarf galaxy]										
	Module					Subarray					
Dithers	B					FULL					
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULEBOX		4		STANDARD				4	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
Special Requirements	1	F115W	F277W	BRIGHT1	4	1	16	16	1202.518		
	Offset 20.0 arcsec, 20.0 arcsec										

Proposal 11908 - Observation 4 - Probing the M-sigma Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs

Observation	Proposal 11908, Observation 4: NIRCAM 3XMM										Tue May 26 23:00:45 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	3XMM_J2150-05_host	RA: 21 50 22.5000 (327.5937500d) Dec: -05 51 9.00 (-5.85250d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F277W	BRIGHT1	4	1	16	16	1202.518		
Special Requirements	Offset 20.0 arcsec, 20.0 arcsec										